

## LAB 2

### Agile Backlog Creation & Sprint Simulation in Jira

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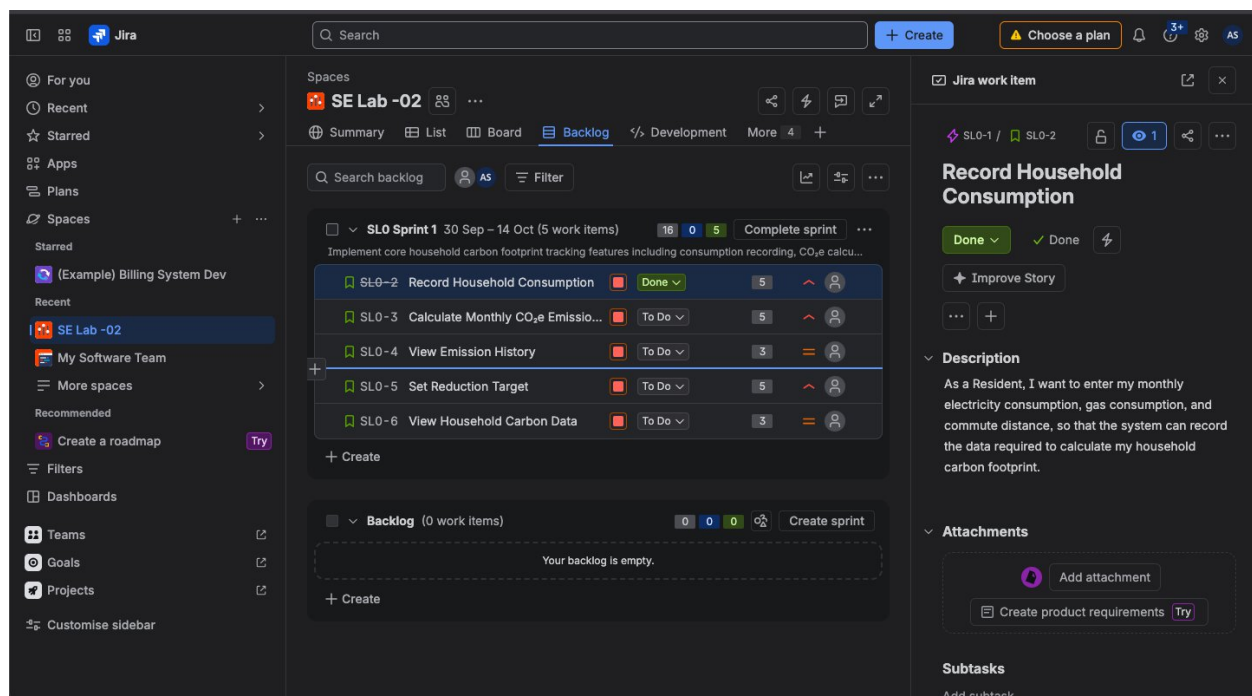
**SRN:** PES1UG25CS803

#### 1. Objective

Creating agile backlog items from functional requirements using Jira. Estimating and prioritizing user stories Creating a sprint simulation Evaluation of sprints using the Sprint Burndown Chart.

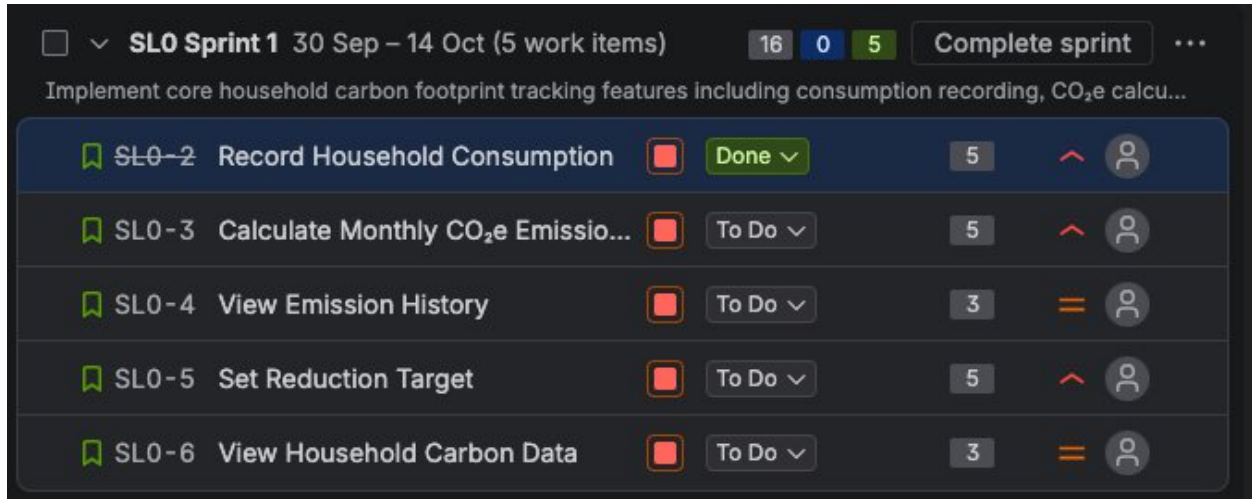
#### 2. Jira Backlog

Six Epics were created for this project along with their associated User Stories to document the major functional requirements of this household carbon footprint system. The first of six of these Epic's is: Epic: Household Consumption Tracking.



### 3. Story Point Assignments

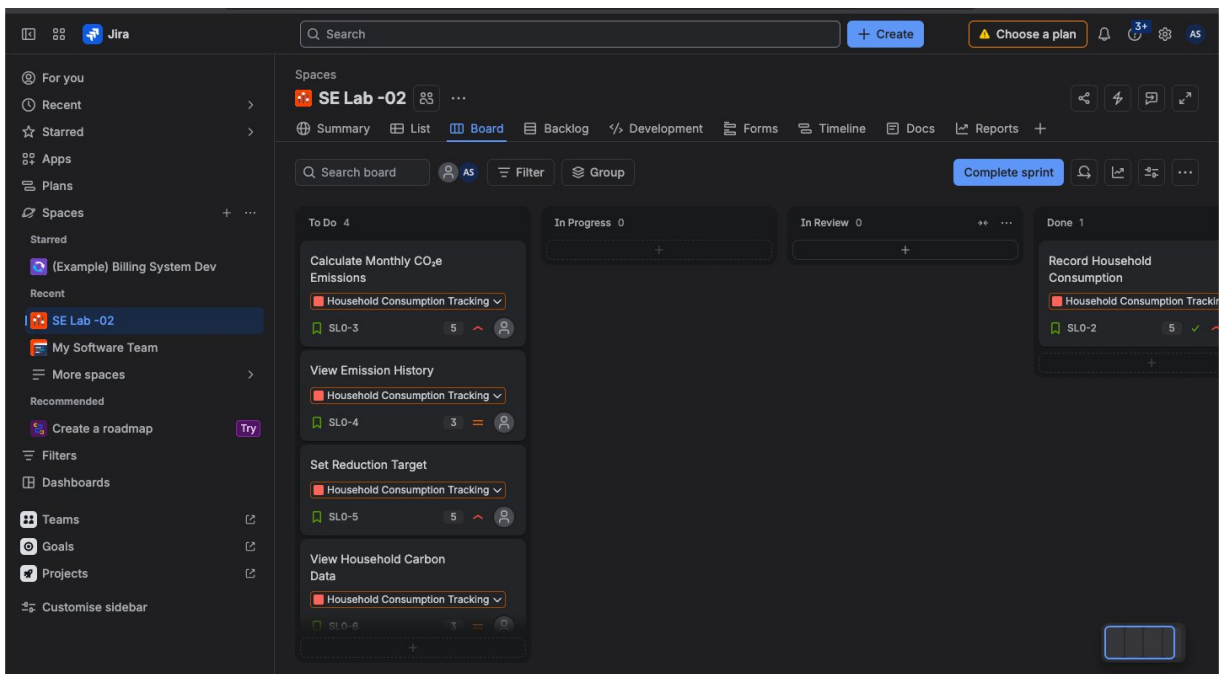
Story points were assigned using the Fibonacci-based estimation approach. The five User Stories were estimated as 5, 5, 3, 5, and 3 story points respectively, giving a total of 21 story points for Sprint 1.



|                                                                                                                         |                                                |                                |    |   |   |                 |     |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|----|---|---|-----------------|-----|
| <input type="checkbox"/>                                                                                                | ▼ SLO Sprint 1                                 | 30 Sep – 14 Oct (5 work items) | 16 | 0 | 5 | Complete sprint | ... |
| Implement core household carbon footprint tracking features including consumption recording, CO <sub>2</sub> e calcu... |                                                |                                |    |   |   |                 |     |
| SL0-2                                                                                                                   | Record Household Consumption                   | Done                           | 5  | ^ |   |                 |     |
| SL0-3                                                                                                                   | Calculate Monthly CO <sub>2</sub> e Emissio... | To Do                          | 5  | ^ |   |                 |     |
| SL0-4                                                                                                                   | View Emission History                          | To Do                          | 3  | = |   |                 |     |
| SL0-5                                                                                                                   | Set Reduction Target                           | To Do                          | 5  | ^ |   |                 |     |
| SL0-6                                                                                                                   | View Household Carbon Data                     | To Do                          | 3  | = |   |                 |     |

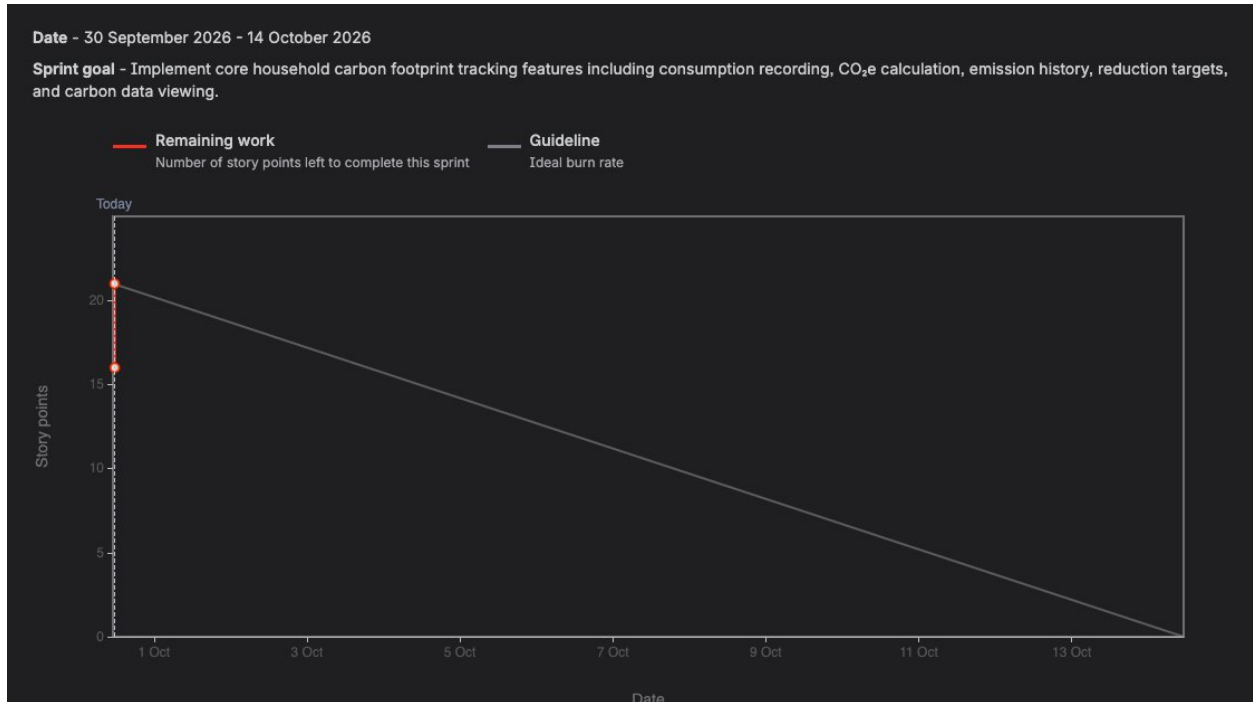
### 4. Sprint Board

Sprint 1 was created in Jira with the five User Stories. The sprint board was used to track the progress of the stories through the workflow stages To Do, In Progress, In Review and Done. During the sprint simulation, the Record Household Consumption story was moved to Done.



## 5. Burndown Chart

The Burndown Chart was used to visualize the remaining story points during the sprint. The sprint initially contained 21 story points. After completing the 5-point Record Household Consumption story, 16 story points remained.



**Work item details**

Search table

| Work Item key | Work type | Work Item priority | Issue summary                                 |
|---------------|-----------|--------------------|-----------------------------------------------|
| SL0-1         | Epic      | Medium             | Household Consumption Tracking                |
| SL0-2         | Story     | High               | Record Household Consumption                  |
| SL0-3         | Story     | High               | Calculate Monthly CO <sub>2</sub> e Emissions |
| SL0-4         | Story     | Medium             | View Emission History                         |
| SL0-5         | Story     | High               | Set Reduction Target                          |
| SL0-6         | Story     | Medium             | View Household Carbon Data                    |

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## **6. Reflection**

### **Q1. Did your estimations reflect the actual effort?**

The estimates were a good approximation for the level of effort from the user's stories. As only a small amount of work was completed during the sprint simulation, the actual amount of work completed during the actual effort was not enough to be useful to make a comparison.

### **Q2. Was your backlog well-prioritized?**

Yes. The backlog was prioritized based on the system functionality. Functionality related to recording the household consumption and the calculation of monthly CO<sub>2</sub>e emissions were a high priority, whereas functionality related to viewing historic and carbon data was a medium priority.

### **Q3. How did your simulated sprint align with your plan?**

The sprint was scheduled with all 5 stories and a total 21 story points. Each story was transitioned across the Jira workflow to emulate how the development would occur. One 5-point story was completed during the simulation and the rest of the stories remained in the To Do phase.

### **Q4. What insights did the burndown chart give about your team's capacity?**

The burndown chart indicated how much work has been finished and the work that was left during the sprint. The overall work at the beginning of the sprint was 21 story points and decreased after being finished a story worth 5 points. This gives you an idea of the team's pace and estimate the amount of work the team can possibly finish in a sprint.

## **7. Conclusion**

The Jira sprint simulation also made clear how functional requirements can be expressed, prioritized, estimated, and controlled within an Agile backlog. The Burndown Chart showed the ongoing status of the sprint's progress and remaining workload.